



MARC
Management

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1.0 Introduction

The mining industry has managed to make safety a main value and create actionable rules to increase safety in operations. Our permanent challenge is to mitigate and better remove the causes of any incidents. Procedures related to maintenance of mining equipment are constantly reviewed and improved due to the risk involved in handling large components. Furthermore, there is also the risk of workers' permanent exposure to crushing, falling and shock hazards, which has the potential to result in energy incidents.

Although there are specific procedures for safely conducting maintenance, the workers conducting these processes should continue to identify additional measures that would mitigate undesired situations.

Another important source of information is the Incident Records of our Clients which helped to develop additional safety measures in order to prevent future incidents from occurring.

In the case of the MD6640 Drill, several operating error incidents with high fatality potential were reported during prolonged maintenance due to uncontrolled movements of the mast machinery (10-ton weight).

2.0 Best Practice Description

This best practice proposes the installation of a hard barrier which prevents any uncontrolled movement or sliding of the mast machinery in the event of operational errors during prolonged maintenance of the MD6640 Drill. The main safety system to fix the drill mast machinery is the transmission pulldown brake applied by a spring when the system is de-energized, and then released electrically when energized. This system involves a risk in the case of involuntary energization due to human error which may release the fixation of the mast machinery.

This best practice proposes the installation of two (2) T/T (thimble-thimble) steel wire rope slings (10 ton each) and two (2) screw pin anchor shackles (8.5 ton each) to secure mast machinery to the mast frame at a certain height. This should prevent any sliding even when brakes are set and ensure that all energy sources are isolated and blocked.

The installation of these slings allows for a hard barrier which guarantees better safety standards for a large-scale work and prolonged exposure to the mast machinery travel area.

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Installation of safety slings to support the mast Machinery	DATE 2/4/2019	CHG NO 00	NUMBER 1.11-180328-03

3.0 Implementation Steps

The implementation of this Best Practice is recommended for prolonged exposure and is targeted to work on mast machinery platform with the mast in vertical position.

Installation of the slings:

Stage	Process
Guidelines	Safety slings should be installed for activities involving prolonged exposure (works involving continuous exposure of at least 4 hours, considering 2 hours for the installation and removal of the slings, which is approx. 50% of the actual work time). This installation procedure should be included in safety documentation and risks should be considered in addition to control measures of the activity.
Use of Slings	Use certified $\frac{3}{4}$ " slings and 1" shackles. Check them frequently for cracks in the slings or forces badly exerted on shackles.
Car positioning	Before starting, when the equipment is still energized, position the upper part of the mast machinery between the 5 th and 6 th diagonal beam of the mast (from top to bottom). The operator shall perform this operation assisted by maintenance personnel.
Brake	The operator must apply the lifting brake and perform the isolation and locking procedure.
Mast anchorage	Safety harnesses must be worn at all times. 2 personnel should conduct this task. Man-lifting equipment must be used for slings and shackles.
Sling Installation	Two $\frac{3}{4}$ " slings must be used in the installation ($\frac{3}{4}$ " as minimum diameter) with 1" shackles (1" as minimum diameter). There must be a sling on each side of the mast machinery, embracing the 5 th diagonal beam and the mast machinery frame. The slings must be tightened as much as possible, requiring 2 people to assist.
Sling Uninstallation	Two people and man-lifting equipment are necessary to perform this operation. With the hoist brake applied and the equipment locked, remove the pin from both shackles, and then remove the shackles. Finally, remove the slings.

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Materials: T/T Slings and Screw pin anchor shackles



Slings installed



Working in the drill mast



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MD6640 Drill Lay-out

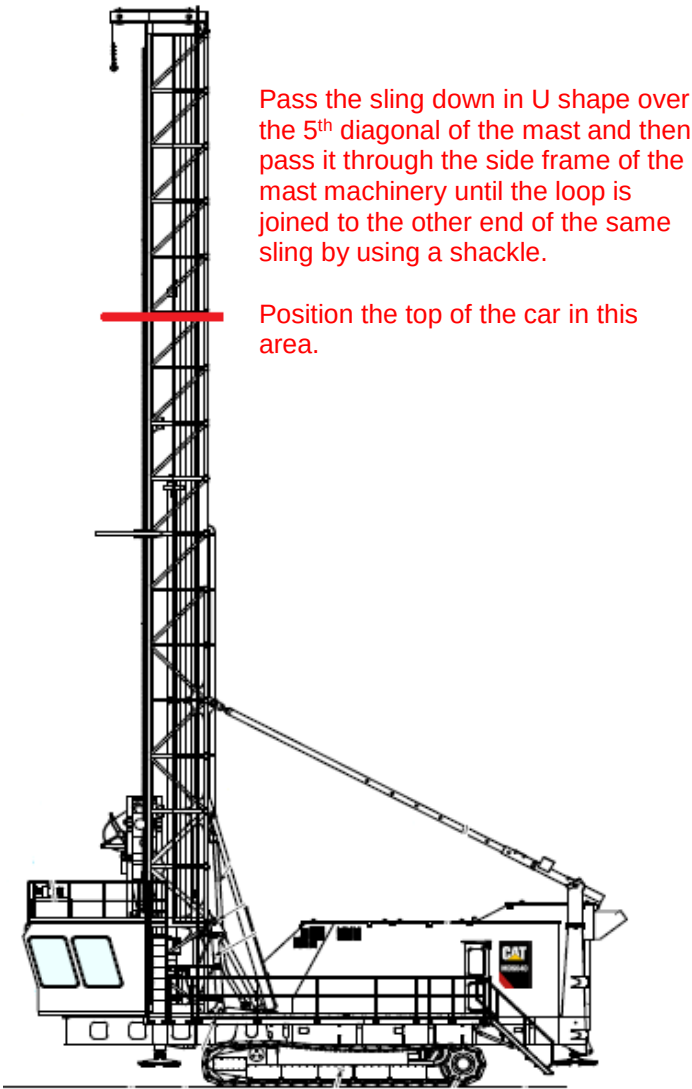


Figure N° 1

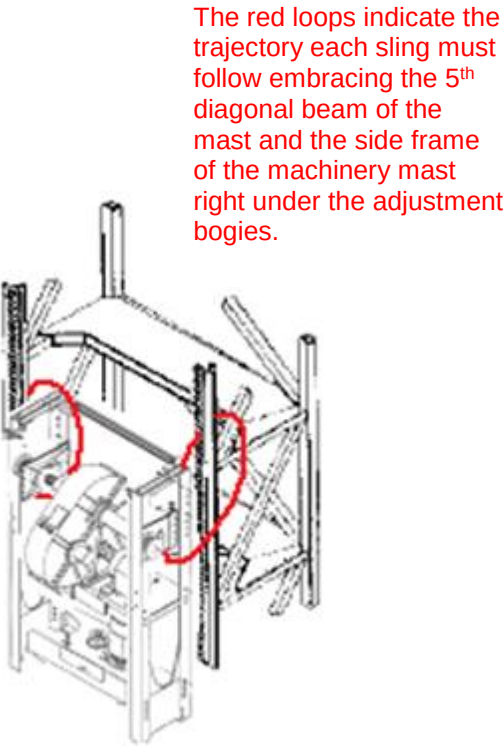


Figure N° 2

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4.0 Benefits

Reduce the possibility of a potentially fatal incident caused by high energy exchange if the mast machinery slides through the mast. In such case, the mast machinery hoist/lifting brake would also fail.

5.0 Resources Required

2 ¾" slings (¾" minimum diameter)
2 1" shackles (1" minimum diameter)
2 mechanics for installation

6.0 Supporting Attachments / References

N/A

7.0 Related Best Practices

N/A

8.0 Acknowledgements

Generation, Implementation and improvement guidelines on Service Report LO04-18

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